**CIS 7000: Multi-Modal AI | Final Project Proposal**

# RL Fine-Tuning of $\pi_{0.5}$ Under Deployment Distribution Shifts

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## 1 Hypothesis

Pretrained vision-language-action (VLA) policies can perform well in their data-collection setting while degrading after deployment shifts. These shifts may be physical and visual differences between a near-replica data-collection site and a production robot, or controlled changes in a simulator. For example, $\pi_{0.5}$ exceeds 0.9 success on the standard LIBERO suites, but falls to 0.20 and 0.17 under LIBERO-PRO position perturbations on LIBERO-Spatial and LIBERO-Object, respectively [1].

We hypothesize that online reinforcement learning with Density Transport (RLDT) can improve $\pi_{0.5}$ success on held-out deployment conditions more sample-efficiently than additional supervised fine-tuning (SFT), while preserving performance in the source setting. The preferred study will examine a real-to-real shift if robot access is available. Otherwise, we will evaluate the same hypothesis using spatial perturbations in LIBERO-PRO. The experimental track will be fixed before adaptation begins rather than selected after observing results.

## 2 Related Work

LIBERO-PRO extends LIBERO with controlled perturbations to objects, initial states, instructions, and environments, exposing substantial generalization failures in current VLAs [1]. The $\pi_{\text{RL}}$ framework introduces Flow-Noise and Flow-SDE formulations for online RL fine-tuning of flow-based VLAs and reports large gains for $\pi_0$ and $\pi_{0.5}$ on LIBERO and ManiSkill [2]. RLDT instead treats policy improvement as transporting an action distribution toward high-reward regions. It constructs a reward-guided transport field and fine-tunes a pretrained flow-matching policy to align with that field without unstable backpropagation through the full generation process [3]. We study whether RLDT can adapt the OpenPI $\pi_{0.5}$ action expert to a held-out deployment distribution rather than only improve the conditions encountered during RL training.

## 3 Study Setup

**Policy and modalities.** We use an OpenPI $\pi_{0.5}$ checkpoint. The policy receives a language instruction, RGB observations, and robot state, and produces robot-action chunks. The real-to-real track will use the checkpoint and camera configuration deployed on the Franka platform. The simulation track will use `pi05_libero`; its released dataset contains 1,693 demonstrations across 40 tasks [4].

**Experimental tracks.** If hardware access is available before training begins, we will study one real manipulation task across a near-replica collection setup and a deployment setup with shifts in lighting, appearance, camera or robot calibration, and hardware behavior. Target-site rollouts will provide task-success rewards; human interventions may additionally provide corrective trajectories for the SFT comparison. If hardware remains unavailable, we will select two to four tasks from LIBERO-Object and LIBERO-Spatial and divide LIBERO-PRO position perturbations into disjoint training and held-out configurations. In either track, the source conditions are retained for measuring performance preservation.

**Methods and objectives.** The primary method is RLDT initialized from the same $\pi_{0.5}$ checkpoint used by all controls. We will freeze the vision-language backbone and update the action expert using RLDT's reward-guided transport objective with a sparse task-completion reward. If the unadapted policy produces too few successes for stable RL, we will use a small SFT warm start and report it as a separate condition rather than silently changing the initialization.

**Evaluation.** The primary metric is task success on held-out target conditions. We will additionally report source-condition success, success as a function of target interactions, training cost, and—for the real track—human-intervention frequency. Each method will use the same condition split and evaluation trials, with multiple training seeds where feasible. Failure rollouts will be categorized as reaching, grasping, or placement failures; explicit recovery training is reserved for future work.

## 4 Controls

We will compare RLDT against: (1) the frozen $\pi_{0.5}$ checkpoint, which measures zero-shot robustness; and (2) SFT from the same checkpoint using successful or human-corrected target trajectories, which tests whether improvement comes from RL rather than simply observing target conditions. All conditions will use the same starting checkpoint and evaluation trials. Where possible, RLDT and SFT will receive matched target-interaction budgets; interaction and demonstration counts will otherwise be reported separately. We will also ablate replay of source demonstrations during RLDT to test whether it prevents degradation in the original setting.

## 5 Expected Outcomes and Contributions

If RLDT improves held-out target success over both the frozen policy and SFT without reducing source success, this would provide evidence that reward-guided adaptation can bridge deployment shifts rather than merely memorize new trajectories. If performance improves only on training conditions, the result would identify overfitting as a limitation of online VLA fine-tuning. If target improvement harms source performance, the replay ablation will quantify the adaptation–retention tradeoff. A null result would reveal whether sparse rewards, insufficient initial success, or limited interactions prevent effective adaptation. The project will contribute a reproducible OpenPI evaluation pipeline, a controlled RL-versus-SFT comparison, and failure analysis under either a real-to-real or simulated deployment shift.

## 6 Limitations and Future Work

Only one experimental track will be executed, so the study will not directly compare real and simulated shifts. The real track may be limited by robot access, safety, and the number of repeatable trials; the simulation track cannot establish transfer to real hardware. VLA inference and online interaction may also limit the number of seeds. Future work will evaluate the same adaptation protocol across both tracks, extend to additional tasks and robots, and test explicit training from recoverable failure states.

## References


[1] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization*. arXiv:2510.03827, 2025. https://arxiv.org/abs/2510.03827.

[2] K. Chen et al. $\pi_{\text{RL}}$*: Online RL Fine-tuning for Flow-based Vision-Language-Action Models*. arXiv:2510.25889, 2025. https://arxiv.org/abs/2510.25889.

[3] B. Lei, K. Daniilidis, and A. Loquercio. *Reinforcement Learning for Flow-Matching Policies with Density Transport*. arXiv:2606.08602, 2026. https://arxiv.org/abs/2606.08602.

[4] Physical Intelligence. *LIBERO LeRobot Dataset*. https://huggingface.co/datasets/physical-intelligence/libero.